

Economics of Climate Change

ECON-4210

Prof. Casey Wichman



Today's goals

- Attendance
- Introduction (👏)
- Overview of the course
 - We'll walk through syllabus, course expectations, etc.
 - Please interrupt with questions.
- For next class...

Why is climate change an economic problem?

- Economists overwhelmingly study market failures
 - E.g., externalities, underprovision of public goods, imperfect competition
- Climate change is the result of a (giant) externality.
 - Market forces won't *internalize* GHGs emissions on their own.
- Greenhouse gas (GHG) mitigation is a public good.
 - Public goods (i.e., GHG mitigation) will be under-provided.
- Environmental/climate economists translate the consequences of climate change into a dollar value (sometimes called the “social cost of carbon”)
- Climate change is a result of incorrect incentives, and economic solutions to climate change aim to correct those incentives.

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Staff members of Rainmaker, a company based in El Segundo, Calif., that uses drones to seed clouds, prepared for a test flight in southern Idaho last year. Alex Goodlett for The New York Times

As the West Dries Up, More Companies Are Promising They Can Make It Rain

States are spending millions of dollars on cloud-seeding technology, backed by new data on its effectiveness — and driven by desperation for water



rees Atlanta Executive discuss the latest reports that the city is Atlanta/Georgia Tech)

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Reuters

Sustainable Business

Norway will drill in Arctic regardless of EU's position, says energy minister

Norway will continue developing its oil and gas resources in the Barents Sea regardless of the European Union's support for a moratorium on Arctic hydrocarbon supplies, and no longer sees itself as Europe's "green battery", Energy Minister Terje Aasland told Reuters.

2 hours ago

WSJ | Buy Side

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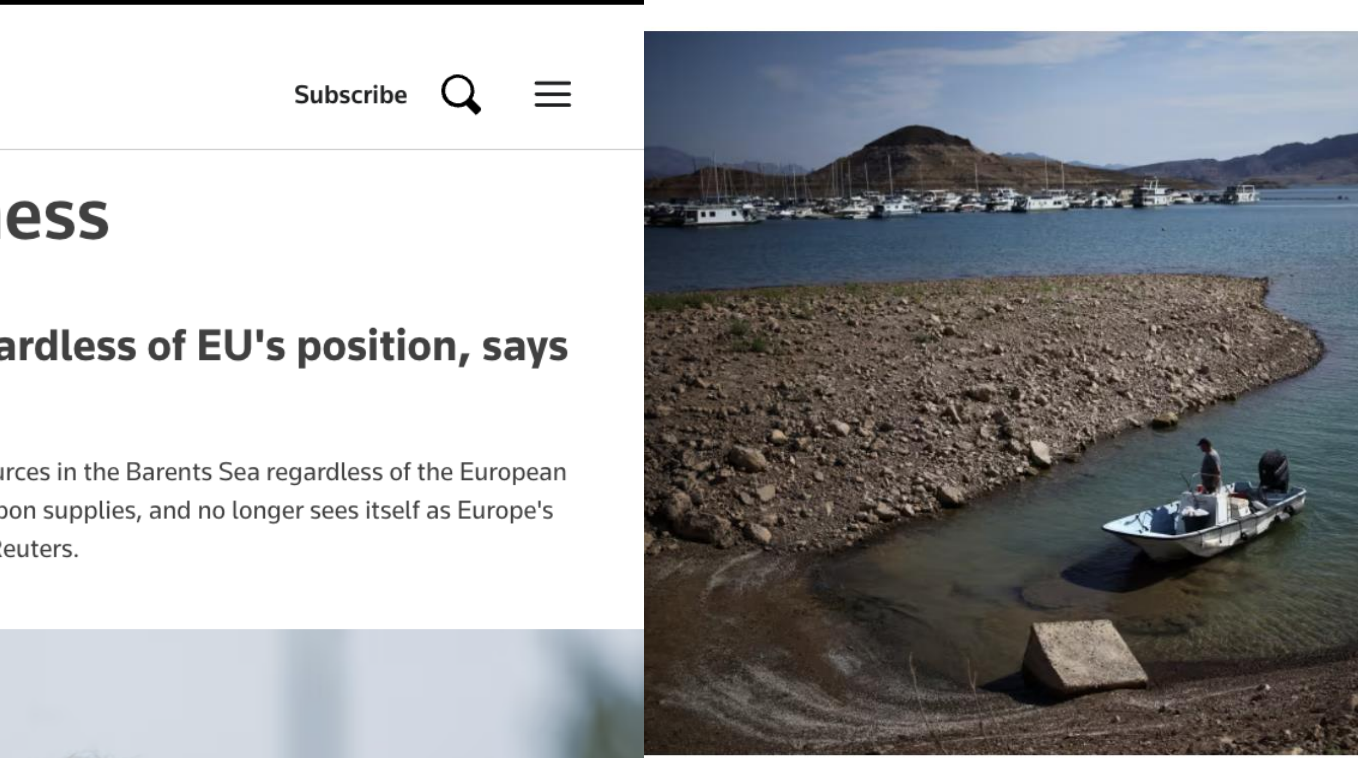


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Two months a forced to exit t that it was blir

Polestar said i that it still doe

By Laura Paddison



The Washington Post

Democracy Dies in Darkness

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The 7

The ocean just broke a temperature record that may have stood for millennia

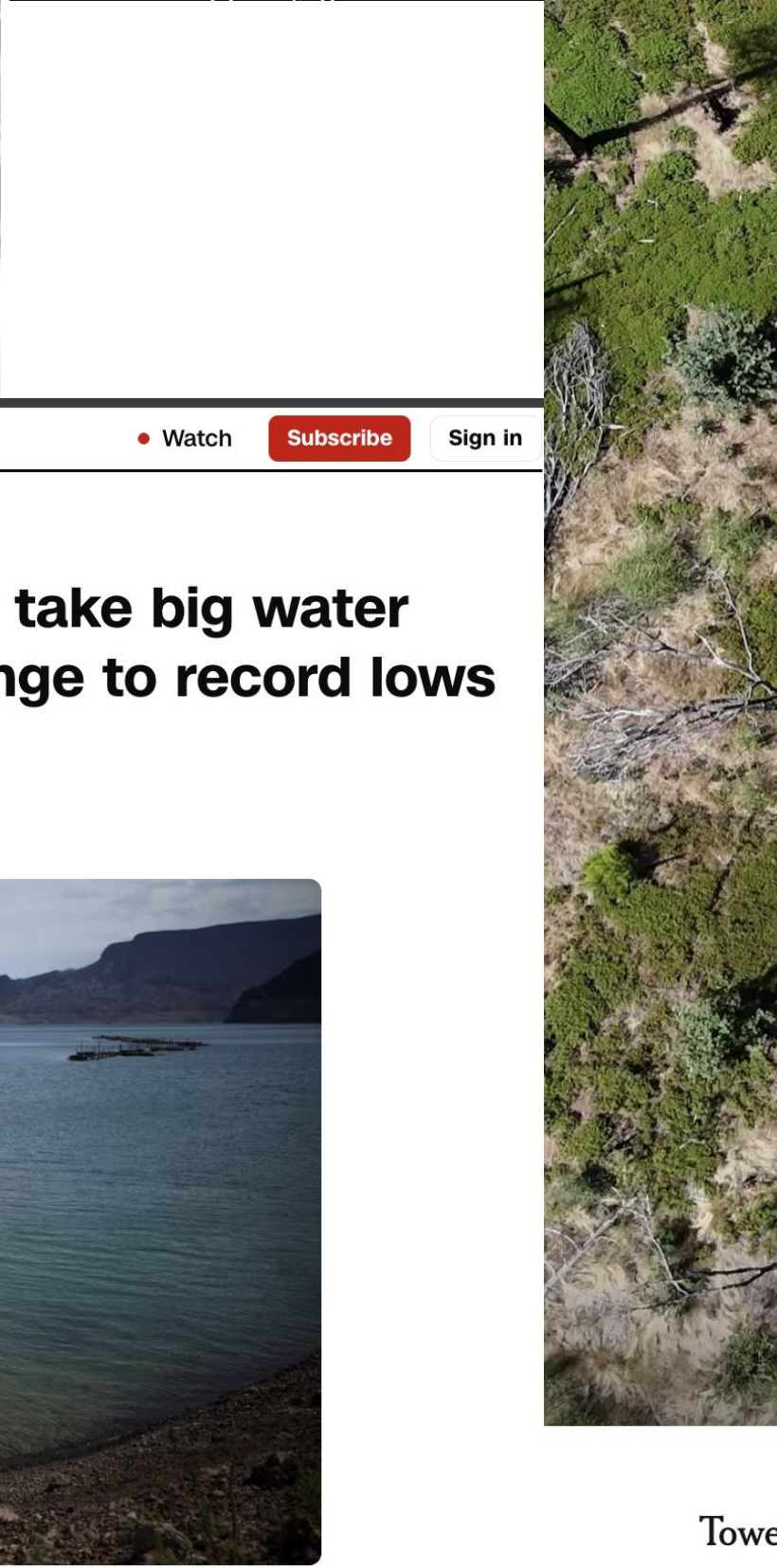
Marine heat waves currently cover 42 percent of the planet's oceans.

Today at 9:08 a.m. EDT

112

By Ben Noll

As a record-breaking El Niño builds in the the globe's oceans are affected by marine h



BBC

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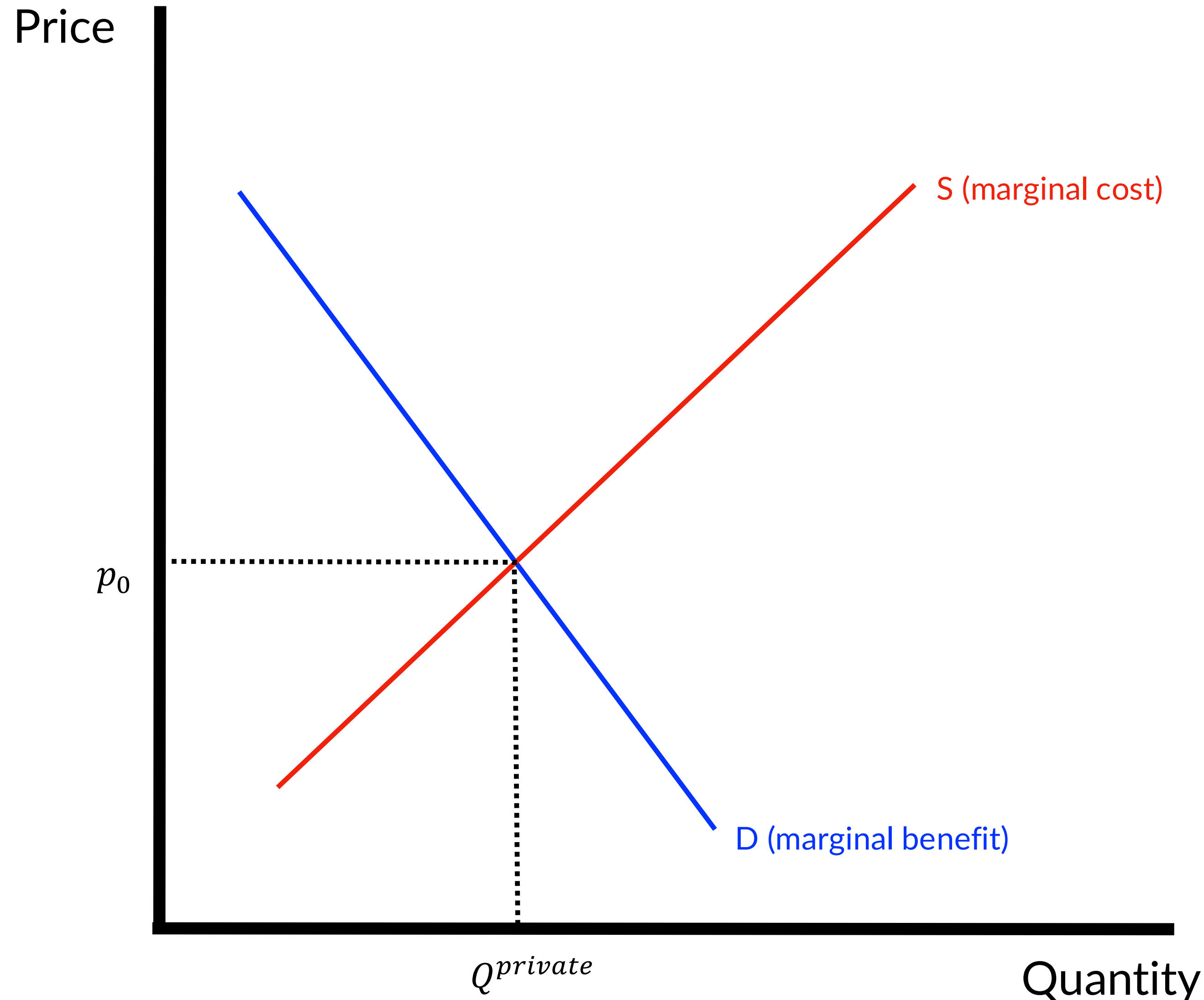
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Tens of thousands urged to evacuate as wildfire approaches Reno, Nevada

Bigger, Hotter Fires Are Slowly Erasing America's Great Forests

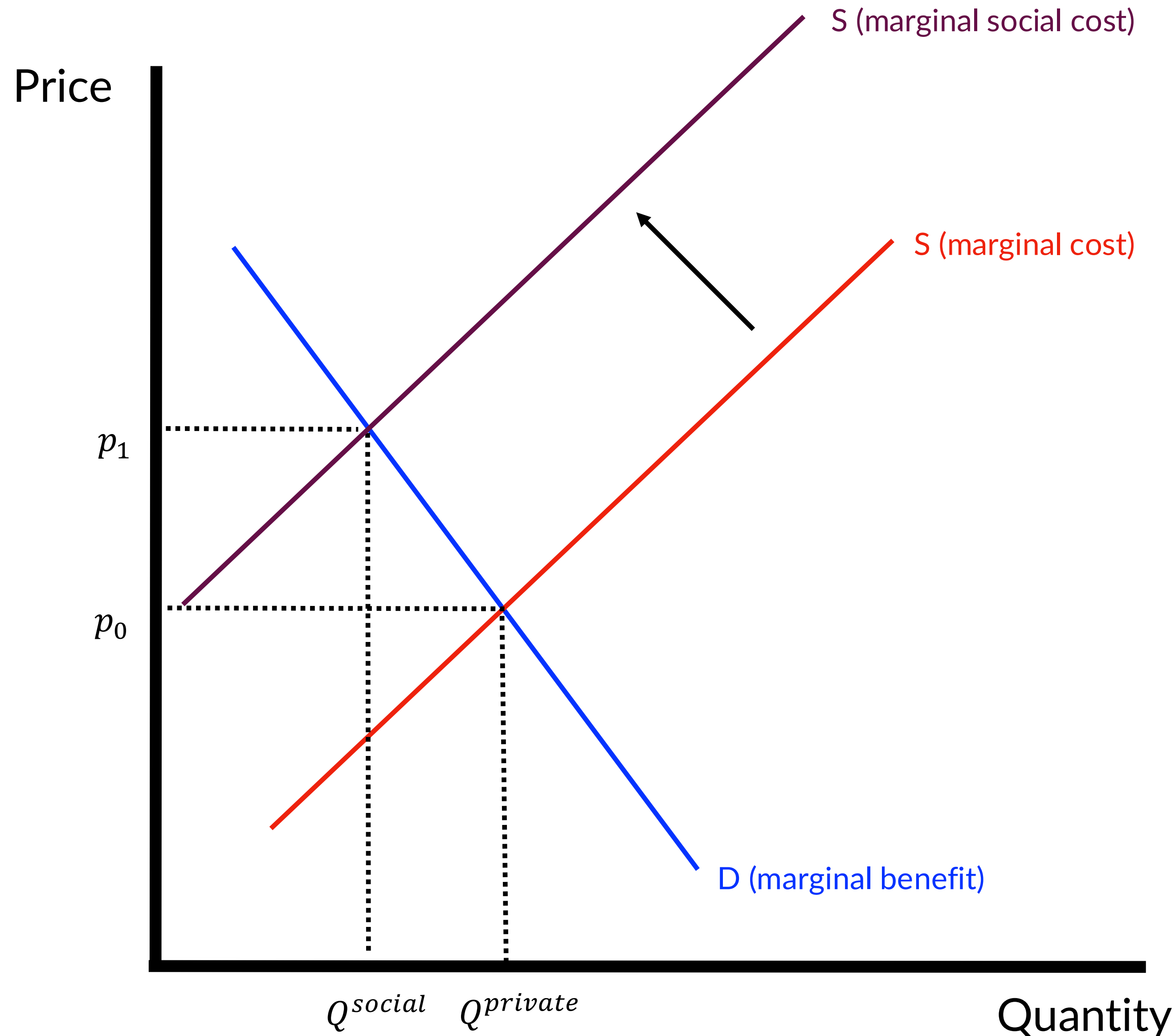
Towering woodlands in the West are being permanently replaced by

The climate problem (from an economics perspective)



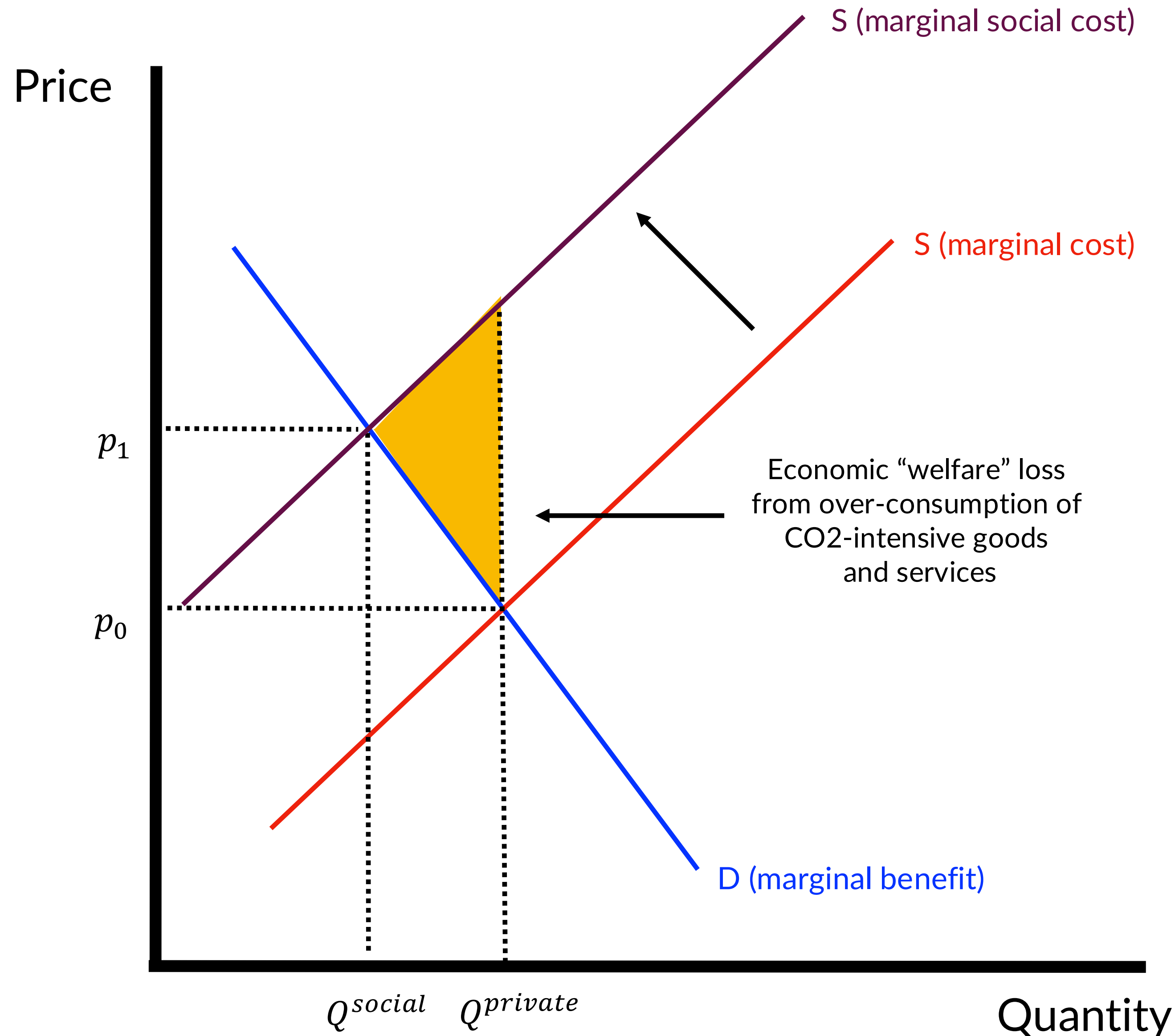
- Think of this as the market for gasoline.
- The benefits we receive from gasoline consumption trace out the demand curve.
 - Higher prices $\rightarrow$ lower quantity demanded.
- The costs of producing gasoline trace out the supply curve.
 - Higher prices $\rightarrow$ higher quantity supplied.
- The equilibrium point balances supply and demand, resulting in market prices (e.g., \$3 per gallon) and quantity demanded (X gallons of gasoline per year).

The climate problem (from an economics perspective)



- Marginal social costs differ from marginal private costs in the case of negative externalities.
- Externalities are costs that are experienced by people “external” to the market.
- Each gallon of gasoline emits ~19.6 pounds of CO_2 when burned
 - CO_2 is a greenhouse gas that contributes to climate change.
 - Other externalities (i.e., local air pollution) arise from tailpipe emissions,
 - Consuming gasoline at p_0 doesn't account for social costs.
- If we paid the full social marginal cost of gasoline, we would pay a higher price (p_1) and consume/produce less (Q^{social}).

The climate problem (from an economics perspective)



- Failure to internalize externalities is justification for market interventions
- Climate change is “simply” a giant negative externality.
- We typically think of climate change as a negative externality that results in overconsumption of CO2-intensive goods and services.
- This course will focus on understanding the nature of this externality and ways to cost-effectively close the gap between privately and socially optimal quantities of CO2-intensive goods and services.

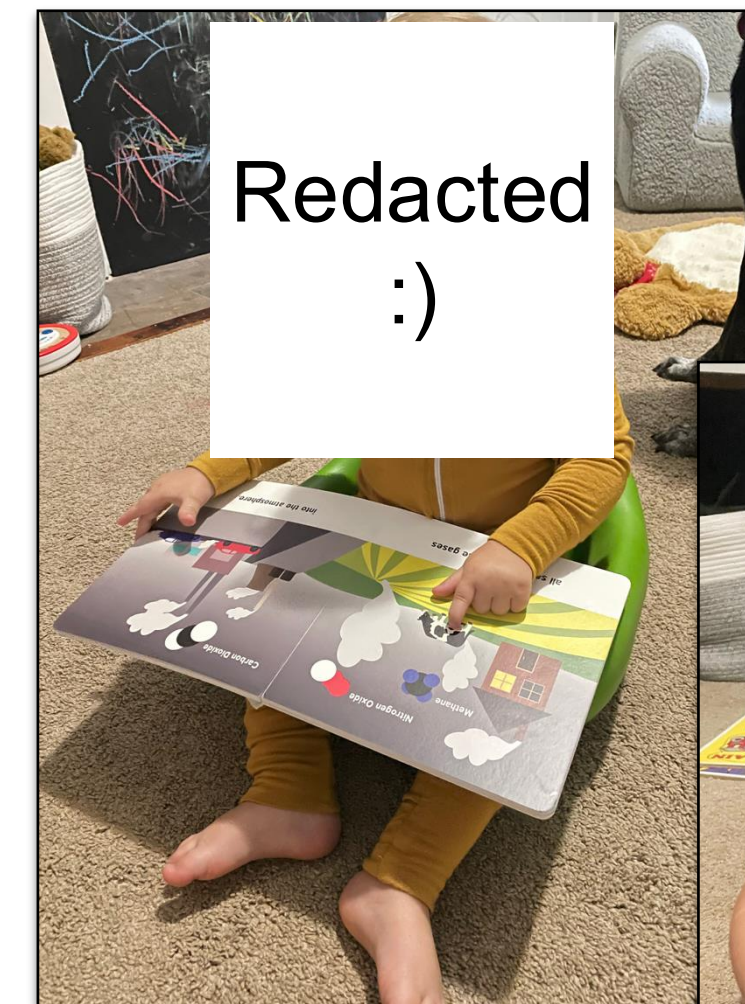
Introduction to Prof. Wichman

About Prof. Wichman

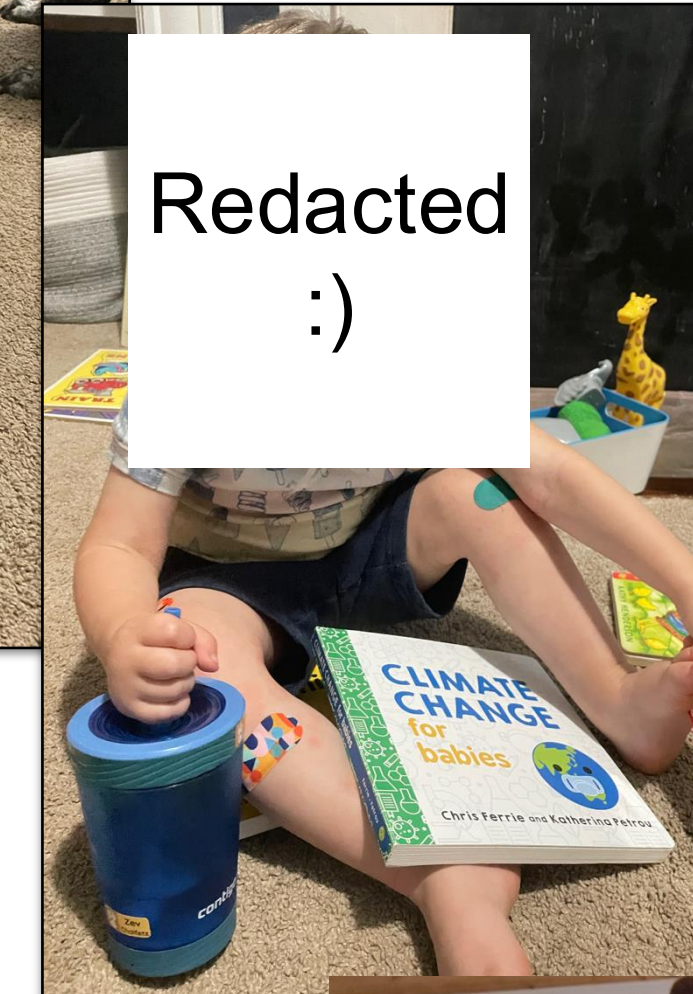
Associate Professor, School of Economics

- Joined Georgia Tech in August 2020.
 - 2019-2020, Research Director at Univ. of Chicago, Energy & Environment Lab
 - 2015-2019, Fellow at Resources for the Future (Env. econ think tank in DC)
 - Originally from Buffalo, NY
- My research focuses on how people interact with the natural and built environment and what that behavior reveals about the value of environmental resources and amenities
 - Household water & energy use, urban transportation, measuring impacts of climate change, outdoor recreation, why people engage in environmentally-friendly behavior
 - Use data/econometrics to estimate causal effects of environmental policy

Fall 2021



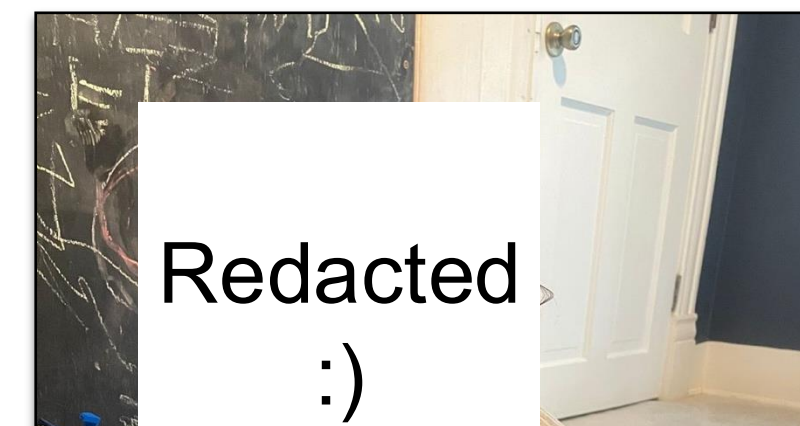
Fall 2022



Fall 2023



Fall 2024



Redacted
:)



Fall 2025



This morning...

Our TA for the semester

- Ana Mazmishvili will be our Teaching Assistant this year.
- Ana is a Ph.D. student in the School of Economics working on environmental and energy topics.
- Feel free to reach out to Ana with questions about the course/materials. She is here to help you understand the material better and succeed in the course.
- Ana will hold virtual office hours each week (**Thursdays, 1-2pm**) and assist me with the course grading, proctoring exams, and so forth.



How this course runs

- **The course website** is home base
 - www.caseyjwichman.com/econ4210 + linked on Canvas
 - Schedule, readings, slides, exam dates.
- **Canvas** for submitting assignments and communicating
 - Problem sets, grades, book chapters, announcements, messages to me.
- **Readings assigned for the class**
 - Journal articles (posted on course website) + select readings from Keohane and Olmstead textbook
- **Help**
 - Me: Mondays 3:30–4:30 PM, Old Civil Engineering 310 (communal office hours).
 - Ana (TA): Thursdays 1:00 PM, virtual — in-person by request.

Fall 2026 at a glance

- **Four units:**

- ▶ 1 **problem set** per unit
 - 5 pts each; 20 pts total
- ▶ 1 **in-class exam** per unit
 - 20 pts each; 60 pts total (drop lowest score)

- **Attendance and participation:**

- ▶ 10 pts for attendance
- ▶ 10 pts for participation in semester long interactive game
 - “Ghost Emissions” (details coming soon!)

Bigger Questions	Detailed Topic
Unit 1: Foundations	
What is climate change?	Climate science 101
How are the climate and the economy linked?	The economy and the climate
Why is climate change an economic problem?	Externalities and public goods
What are the costs and benefits of climate policy?	Benefit-cost analysis
Unit 2: Discounting and climate damages	
Should I care about my grandchildren’s future?	Discounting benefits and costs
Can we put a price on climate impacts?	Measuring/valuing climate impacts and the social cost of carbon
Who bears the burden of climate impacts?	Equity & climate impacts
Unit 3: Climate policy	
How would an economist solve climate change?	Optimal climate policy
Who pays for a carbon tax?	Equity & climate policy
Does behavior undermine climate policy?	Unintended consequences
How do we cut emissions cheaply?	Innovation & technological change
Unit 4: Adaptation, uncertainty, and cooperation	
Can’t we just live with climate change?	Adapting to climate change
What don’t we know about climate change?	Uncertainty, fat tails, & tipping points
Can we fix climate change with this weird trick?	Geoengineering & offsets
Can we address climate change on our own?	International cooperation
Is changing our own behavior the solution?	Behavioral change

Course website + syllabus

www.caseyjwichman.com/econ4210

For next class...